

# Alexander Hasenbein

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Built a self-driving RC car with object detection at 13, an autonomous drone implementing Coast Guard Victor Sierra (sector) search patterns at 14, hand-built a laptop from individual components, and developed a working blockchain in high school.

## EDUCATION

### The Pennsylvania State University

B.S. Computer Science, B.S. Mathematics | GPA: 3.5

University Park, PA

Expected May 2028

Commonwealth Engineering Scholarship • Behrend Honors Program • Nittany AI Leadership Academy • ELP

## WORK EXPERIENCE

### Flex

Austin, TX

Software Engineering and Machine Learning Intern

May 2026 - Aug 2026

- Built and deployed an ML pipeline in Python (**LightGBM, pandas, scikit-learn**) processing **1M+ records** to forecast throughput, achieving 90% accuracy on predictions within a 30 second window of build time.
- Engineered features from SQL Server views and optimized complex queries across production datasets of **4M+ rows**, implementing parallelized chunking to eliminate tempdb exhaustion.
- Designed web-based dashboards in C# (ASP.NET) with dynamic charts, tables, and filters, delivering real-time analytics for the process engineering team. Owned delivery of 2 HQ-wide projects, managing version control and roadmap tracking.
- Built data analysis apps in Python using Plotly and Pandas, reducing analysis time by 10% against baseline (standard tools).

### The Pennsylvania State University

Erie, PA

Undergraduate Researcher, Honors Option

Jan 2025 - May 2025

- Built a hyperparameter optimization framework for PyTorch neural networks predicting quadratic roots, benchmarking Bayesian optimization (Optuna), random search, and multi-objective pareto search to quantify accuracy-efficiency tradeoffs.
- Built a data-pipeline and full evaluation suite (MAE, RMSE,  $R^2$ , per class-error, inference speed) with interactive dashboards for hyperparameter analysis, trial exploration, ensuring reproducible, interpretable results.

## PROJECTS & HACKATHONS

### DevCord – Multi Agent AI Orchestration Platform | 1<sup>st</sup> Place - AI Agents & LLMs: Behrend Hackathon

TypeScript, Discord.js, OpenRouter

- Architected specialized agents (Manager, Dev, QA, Command) with scoped roles and task handoff in strict TypeScript using Discord.js and OpenRouter, with modular model routing to swap LLMs per agent without modifying orchestration logic.
- Implemented autonomous project kickoff, thread-based task decomposition, persistent run/state tracking with pause/resume/cancel controls, plus voice-first Q&A via local Whisper STT/TTS in Discord voice channels.

### Particles Interlinked

Next.js, React, Three.js, Node.js

- Developed a complex real-time particle simulation system using Three.js with WebGL rendering, applying physics concepts such as particle interactions, gravitational attraction, and collision detection implemented efficiently in JavaScript.
- Integrated performance monitoring and leveraged WebGL for hardware-accelerated graphics rendering, demonstrating strong skills in performance tuning and UI/UX for interactive simulations, **increasing fps by ~22% consistently**.
- Migrated from a 2D to a 3D library, improving visual dynamism and **rendering speed** over the prior implementation.

### BowieForum

React, MongoDB, SQL, Node.js

- Designed and deployed a student forum platform with 150+ users. Utilized optimized SQL queries to reduce server load.
- Used pagination to efficiently load posts to reduce strain both client side and server side while keeping low latency.
- Developed React frontend, Node.js backend, and integrated into MongoDB Database. Deployed with a DevOps plan.

## SKILLS

- Programming Languages:** Java, C++, C#, JavaScript, TypeScript, Python, SQL, VB.NET, HTML, CSS
- Frameworks:** React, Next.js, Express.js, Spring, ASP.NET, FastAPI, Flask, LightGBM, pandas, scikit-learn, Plotly, NumPy
- Tools & Technologies:** Git, Docker, MongoDB, SQL Servers, Node.js, OpenRouter, CI/CD Pipelines, AWS, Azure